

### **Slide 1 – Title**

My name is Veera Saiva Sai Lakshmi Kalyan, and I'm here with Taran Mamidala and Jupalli Sai Abhisekh Reddy. We're excited to share our work. Our project is on **“Enhanced Image Inpainting using Transformer-GAN Hybrid Models”**, and we'll walk you through it.

**Slide 2 – Problem Statement** When you look at an old photograph with scratches or try to remove an unwanted object from a selfie, you want the final picture to look seamless—as if nothing was missing or damaged. In many existing methods, the repaired region either looks blurry, fails to respect the original structure, or visibly mismatches the surrounding pixels.

So Our aim was to build a system that reliably restores missing regions with sharp textures, accurate geometry, and smooth blending so that even a trained eye can't tell where the image was inpainted.

### **Slide 3 – Introduction to Image Inpainting**

Image inpainting is simply the process of filling in holes or erasures in a picture so that the repaired image appears natural. Beyond restoring old photos, it powers modern photo-editing tools to remove objects from landscapes or people from group shots, and it even plays a role in medical imaging by reconstructing occluded or noisy scan regions. By teaching a model to predict what should lie behind a masked area, we can automate a task that would otherwise require skilled manual editing.

**Slide 4 – Dataset & Preprocessing** We chose the CelebA dataset, which offers over 200,000 aligned and cropped celebrity face images, each annotated with 40 binary attributes. To evaluate fairly, we followed the standard split—162,770 images for training, 19,867 for validation, and 19,962 for testing. During preprocessing, we normalized pixel values to the  $-1$  to  $1$  range and applied a random  $56 \times 56$ -pixel square mask to each training image. This creates paired examples of “input” (masked image) and “ground truth” (original image), enabling the model to learn exactly what's missing.

### **Slide 5 – GAN Basics**

our approach was to use Generative Adversarial Networks, or GANs. A GAN has two networks against each other: a generator—which tries to create realistic inpainted images—and a discriminator—which tries to distinguish real images from the generator's output. Through this adversarial training, the generator learns to produce ever-more convincing textures and structures, improvise by the discriminator's feedback whenever it spots a flaw.

### **Slide 6 – Our Training Strategy**

For each of our three architectures, we combined an adversarial loss (so the generator tries to fool the discriminator) with a reconstruction loss (L1 pixel-wise error so it matches the ground truth) and, in our hybrid model, a perceptual LPIPS loss (so high-level features

align). By balancing these, we ensure the inpainted patch is both visually sharp and semantically consistent with the rest of the face.

**Slide 7 – Related Work & Baselines** Before diving into our designs, we looked at two main lines of prior work. First, classic inpainting methods like *PatchMatch* (Barnes et al., 2009) are fast but often miss semantic context, and *Contextual Attention GAN* (Yu et al., 2018) introduced learnable attention to focus on missing regions. Second, transformer-based restorers like *IPT* (Chen et al., 2021) and *SwinIR* (Liang et al., 2021) show great performance on denoising and super-resolution but can be heavy to train. Our hybrid approach marries a U-Net generator and spectral-normalized PatchGAN discriminator with optional LPIPS loss, leveraging convolutional inductive bias alongside adversarial and perceptual training.

**Slide 8 – Three Architectures Implemented** We built and compared:

1. a **DCGAN-based generator** with explicit mask processing,
2. a **BigGAN-inspired residual generator** outfitted with self-attention modules, and
3. our **Transformer-GAN hybrid**, which concatenates masked image and mask channels into a U-Net backbone, uses spectral normalization, and adds LPIPS perceptual loss. Each model was trained under identical settings so we could fairly judge their strengths and weaknesses.

**Slide 9 – Quantitative Results**

On our held-out test set (19,962 images), we measured three metrics:

- **L1 Loss** (mean absolute pixel error)
- **PSNR** (peak signal-to-noise ratio, in dB)
- **SSIM** (structural similarity index)

| Model                                | L1 Loss ↓    | PSNR ↑ (dB)  | SSIM ↑      |
|--------------------------------------|--------------|--------------|-------------|
| DCGAN-Based                          | 0.042        | 22.0         | 0.80        |
| BigGAN-Inspired Residual + Attention | 0.035        | 23.5         | 0.84        |
| <b>Transformer-GAN Hybrid</b>        | <b>0.030</b> | <b>25.09</b> | <b>0.86</b> |

Our hybrid model clearly outperformed both baselines on every metric, validating the benefits of combining global context with adversarial and perceptual training.

**Slide 10 – Training Curves** Here you can see

On the left, the L1 Loss histogram shows that most of our inpainted patches fall between 0.02 and 0.03, with a long tail extending out toward 0.05. This tight clustering around a low error confirms that—even when we randomly mask out different 56×56 regions—the model consistently reconstructs missing pixels with minimal absolute error. In the center, the PSNR distribution peaks around 25 dB and ranges roughly from 22 dB up to 30 dB for the

best reconstructions. This indicates strong denoising and sharpness retention: higher PSNR values correlate with fewer pixel-level discrepancies.

Finally on the right, the SSIM scores are overwhelmingly above 0.85, with many reconstructions scoring over 0.9. This skew toward high SSIM tells us the model preserves structural and perceptual similarity exceptionally well, reinforcing that the hybrid architecture captures both global context and fine details.

the loss trajectories across 10 epochs. The discriminator loss settles around 0.6–0.7, indicating it remains a challenging adversary, while the generator loss steadily declines—showing stable adversarial training without mode collapse. We used a cosine annealing scheduler to gently reduce learning rates, which smoothed out late-epoch oscillations.

### **Slide 11 – Qualitative Comparisons —**

DCGAN model often left soft, blurry patches. The BigGAN- network sharpened details but sometimes misaligned facial features in larger masks. In comparison with, our hybrid model consistently produced crisp textures and correct geometry, whether filling in missing eyebrows or restoring lips, with the inpainted regions seamlessly matching surrounding skin tone and lighting.

Here we plot the discriminator and generator losses over five epochs. The discriminator starts around 0.7, dips to about 0.55 by epoch 4—showing it remains a challenging. The generator loss begins near 2.8 in epoch 1, drops to about 2.35 by epoch 2, then gradually rises to 2.5 before settling around 2.45. Those early gains reflect rapid learning of basic inpainting features; the gentle uptick later on shows the adversarial.

Overall, the smooth curves and absence of large oscillations or collapse indicate stable training—thanks in part to our cosine-annealing learning-rate schedule and the patch-based adversarial setup.

**Slide 12 – Key Takeaways** To sum up, combining transformer-style global context with GAN-based detail generation—and reinforcing it with a perceptual loss—yielded the strongest performance. Our Transformer-GAN hybrid outperformed both DCGAN and BigGAN baselines across every metric, offered stable training, and delivered the most natural looking inpainted faces.

### **Slide 13 –**

**Future Work** Looking ahead, we're excited to explore different mask shapes and sizes so the model adapts to arbitrary missing regions, pre-train on even larger and more varied datasets for broader generalization, and extend the approach to video inpainting where maintaining temporal consistency is crucial.

Thank you! We hope you found our approach and results enlightening, and we're happy to answer any questions you may have.

Here are some of the key challenges we faced in this inpainting project:

**1. Random Mask Variability**

Because we applied square masks at random locations for each image, the model had to learn to recover a huge variety of missing regions—from facial features like eyes and mouths to background areas. Ensuring robust performance across all possible masks required extensive data augmentation and careful tuning of mask size and placement.

**2. Balancing Adversarial and Reconstruction Losses**

In our hybrid GAN-based setups, we combined pixel-wise (L1) loss, adversarial loss, and perceptual (LPIPS) loss. Finding the right weights ( $\lambda$  values) so that the generator didn't overfit to pixel accuracy at the expense of realism—or vice versa—took many rounds of experimentation.

**3. Training Stability and Mode Collapse**

GAN training is famously unstable. We saw occasional mode collapse (the generator producing nearly identical patches) when the discriminator became too strong. To mitigate this, we used techniques like spectral normalization, patch-based discrimination, and cosine-annealing learning-rate schedules—but fine-tuning remained time-consuming.

**4. Computational Resources and Training Time**

Working with 162K+ images at 224×224 resolution, training three different models (DCGAN, BigGAN-inspired, Transformer-GAN hybrid) for multiple epochs demanded hundreds of GPU hours. Managing batch sizes, number of workers, and data-loading bottlenecks was critical to keep iterations moving.

**5. Semantic Coherence vs. Local Detail**

Models like PatchMatch can fill holes quickly but lack high-level understanding, while pure transformer approaches capture semantics but struggle with sharp edges. Our hybrid architectures improved on this, yet striking the right balance—so that fillings both “make sense” and “look real”—remains a core challenge.

**6. Evaluation Metric Limitations**

PSNR and SSIM quantify pixel-level fidelity and structural similarity, but they don't always correlate with perceptual quality. We still observed cases where a high SSIM mask looked slightly “blurry” to the human eye. Developing or adopting better perceptual metrics (e.g., FID, user studies) is an open area.

7. **Generalization to Irregular Masks**

We trained on square masks of fixed size, but real-world applications often involve irregular or patterned occlusions. Extending the model to handle arbitrary mask shapes—and ensuring it doesn’t “cheat” by learning mask priors—will require more diverse masking strategies during training.

8. **Attribute Imbalance in CelebA**

Some facial attributes (e.g., “Bangs,” “Eyeglasses”) appear in only a small fraction of images, making it harder for the model to learn to inpaint those specific features convincingly when they’re occluded. Addressing this imbalance could involve targeted oversampling or curriculum learning.